

EESHWAR ET

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SUMMARY

Result-oriented ECE student with hands-on experience on embedded systems and integrating sensors with Arduino, Raspberry Pi, STM32 platforms. Proficient in Embedded C, Python, and using debugging tool like VS Code. Seeking an Entry-Level Embedded Engineer position to apply skills in hardware-software interface optimization and drive product reliability.

EXPERIENCE

- Designed a dual-microcontroller environmental monitoring system using Arduino Uno and ESP32. Implemented I²C for local LCD display and UART for reliable communication between controllers. Integrated temperature, humidity, soil moisture, and ambient light sensors to provide continuous monitoring and cloud visualization. The system reduces manual inspection, enables real-time remote monitoring, improves response time to abnormal conditions, and supports preventive maintenance through threshold-based alerts.

PROJECT

Smart Environmental Monitoring System

- Designed a dual-microcontroller environmental monitoring system using Arduino Uno and ESP32. Implemented I²C for local LCD display and UART for reliable communication between controllers.
- Integrated temperature, humidity, soil moisture, and ambient light sensors to provide continuous monitoring and cloud visualization.
- The system reduces manual inspection, enables real-time remote monitoring, improves response time to abnormal conditions, and supports preventive maintenance through threshold-based alerts.

Industrial Motor Monitoring And Protection System

- Developed an industrial motor protection system on STM32 Nucleo-L031K6 using PWM, hardware timers, ADC, UART, and interrupts.
- Implemented real-time fault detection, automatic motor shutdown during abnormal conditions, and live status monitoring.
- The design reduces equipment damage risk and demonstrates industrial-grade embedded firmware development.

EDUCATION

B.E in Electronics and Communication Engineering

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CERTIFICATIONS

Embedded systems

- intermediate.

SKILLS

MICROCONTROLLERS: STM32 NUCLEO, RASPBERRY PI PICO, ESP32, ARDUINO UNO

COMMUNICATION PROTOCOLS: SPI, I2C, UART

PROGRAMMING LANGUAGES: EMBEDDED C, PYTHON

TOOLS: VS CODE
